# Prüfer Code-Based Procedural Dungeon Generation:

### Theory, Implementation, and Game Application

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## Abstract

This project explores the application of Prüfer codes—a bijective encoding of labeled trees—to procedural dungeon generation in games. While traditional approaches employ ad hoc algorithms (Random Walks, Binary Space Partitioning, Delaunay Triangulation + MST), this work demonstrates that Prüfer codes provide a theoretically elegant, memory-efficient, and computationally efficient alternative for generating tree-structured dungeon layouts.

**Key Findings:** Prüfer codes excel at large-scale map generation (O(n log n) average, O(n) space) and offer systematic graph control, but are topologically limited—all rooms are adjacent (no multi-hop corridors)—making them suitable for abstract adjacency structures rather than spatially-aware dungeons.

**Contributions:**

- Memoryless O(n log n) Prüfer decoding eliminating full graph storage
- Deterministic biome assignment using graph metrics (degree, depth, eccentricity, centrality)
- Interactive Prüfer code visualizer with animation and code input
- Comparative complexity analysis against Delaunay + MST and BSP methods

# 1. Introduction

## 1. 1 Background and Motivation

Procedural content generation (PCG) is a process of game generation well used in modern game development. It enables developers to create vast, diverse, and replayable environments. Among the structures most frequently generated procedurally in games are tree-like maps and dungeon layouts, which appear in diverse contexts:

- **Dungeon crawlers** (e.g., Pixel Dungeon, Spelunky, Binding of Isaac): Roguelike games require fast generation of interconnected room layouts with no dead ends and varied connectivity
- **Branching narratives:** Quest trees, dialogue trees, and skill progression systems often use tree structures
- **Tournament brackets:** Competitive game structures form perfect binary trees

Traditional procedural dungeon generation employs several well-established algorithms [1] :

| Algorithm | Approach | Key Features | Limitations |
|---|---|---|---|
| **Random Walk** | Simulate wandering path on grid | Fast, simple, generates mazes | Can create dead ends, unpredictable connectivity |
| **Binary Space Partitioning (BSP)** | Recursively subdivide space | Predictable, hierarchical, non-overlapping rooms | Rigid rectangular structure, limited variety |
| **Delaunay Triangulation + MST** | Spatial clustering + graph algorithms | Optimal corridor length, natural room distribution | Requires geometric computations, implementation complexity |

While effective, these methods are often used **ad hoc**, lacking a unified theoretical foundation, making it difficult to analyze them. Additionally, they require explicit storage of full geometric or adjacency information.

**Research Question:** Can Prüfer codes, a classical construct from combinatorial graph theory, provide a more systematic, theoretically robust, and computationally efficient foundation for procedural tree- based map generation?

## 1.2 Prüfer Codes: Mathematical Foundation

### Cayley's Formula and Enumeration

Cayley's formula states that the number of labeled trees on n vertices is exactly n^(n-2). This remarkable result immediately raises a practical question: how do we efficiently enumerate or sample from these n^(n-2) possible trees?

The answer: **Prüfer codes**, which establish a bijection (one-to-one correspondence) between all labeled trees with n vertices and all sequences of n-2 integers in the range [1, n].

**Bijection:** Given a labeled tree T on n vertices, its Prüfer code P is constructed as follows:

1. Repeatedly find the leaf (degree-1 node) with the smallest label
2. Record the label of its unique neighbor
3. Remove the leaf from the tree
4. Repeat n-2 times

The final two remaining vertices are recorded as the last edge.

**Reverse Process:** Given a Prüfer code P, the original tree can be uniquely reconstructed by:

1. Computing the degree of each vertex (degree[v] = 1 + count of v in P)
2. Iteratively finding the smallest-labeled leaf and connecting it to the corresponding element of P
3. Decrementing degrees and reinserting newly-created leaves into a sorted structure
4. Connecting the final two degree-1 vertices

**Example:**

```
Prüfer Code: [2, 3, 2] (remove leaves 1, 4, 3 in that
order)
Reconstruction from [2, 3]:
- Degrees: [1,3,2,1,1] (vertex 2 and 3 appear once each)
- Step 1: Smallest leaf=1, connect to 2,
  degrees=[0,2,2,1,1]
- Step 2: Smallest leaf=4, connect to 3,
  degrees=[0,2,1,0,1]
- Step 3: Smallest leaf=3, connect to 2,
  degrees=[0,1,0,0,1]
- Step 4: Connect remaining leaves 2,5
```

```
Tree:    1-2-3
           | |
           5 4
```

### 1.3 Why Prüfer Codes for Game Design?

1. **Theoretical Elegance:** Bijective encoding enables sampling uniformly from all n^(n-2) possible trees
2. **Memory Efficiency:** A Prüfer code requires only O(n) space (a sequence of n-2 integers) versus O(n) for full adjacency lists
3. **Deterministic Reproducibility:** The same Prüfer code always generates the same tree, enabling seeded world generation
4. **Computational Efficiency:** O(n log n) typical decoding complexity (O(n) optimal with more space complexity implementation)

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## 2. Literature Review

### 2.1 Prüfer Codes: Foundations and Applications

**Fundamental Work:**

Prüfer codes were first introduced as a mechanism for proving Cayley's formula. The bijection between trees and sequences has since been extended and refined. Caminiti et al.[3] provide a unified framework for tree coding schemes, including optimal linear-time encoders and decoders.

**Key Algorithmic Results:**

- **O(n log n) Decoding** with priority queues (standard approach)
- **O(n) Decoding** using pointer-based leaf tracking[3] (employed in visual and game)
- **O(n) Encoding** with similar pointer techniques

### 2.2 Procedural Dungeon Generation

**Classical Approaches:**

**a) Delaunay Triangulation + MST:**

The industry-standard approach combines:

1. Random room placement
2. Delaunay triangulation of room centers (guarantees no "bad" triangles)
3. MST computation (Prim's or Kruskal's, O(n log n))
4. Random edge reinsertion for connectivity loops

Advantages include natural spatial distribution and minimal corridor length. Disadvantages: requires geometric computation, complex implementation, O(n log n) time for both triangulation and MST

**b) Binary Space Partitioning (BSP):**

Recursively subdivides a rectangular space, placing rooms in leaf nodes [1] . O(n) time complexity, but generates rigid rectangular structures with limited variety.

**c) Random Walk / DFS (employed for random/deterministic traversal):**

Simple, fast (O(n)), but prone to dead ends and low connectivity without post-processing.

### 2.3 Game Design and Biome Systems

**Current Gap:** While Prüfer codes are theoretically elegant, their application in procedural game generation is underexplored. Existing work focuses on bottom-up spatial methods (Delaunay) or ad hoc algorithms (BSP, Random Walk). No prior work systematically applies Prüfer codes to dungeon generation.

**Work Contribution:** This project demonstrates that:

- Prüfer codes enable systematic, efficient tree-based dungeon generation
- Graph-theoretic properties can be mapped to game environment design
- Deterministic seeding makes reproducible, large graph based worlds
- Trade-offs exist: Prüfer excels at large-scale generation but lacks spatial optimization

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## 3. Formal Problem Statement of Project

**Input:** An integer n (number of rooms)

**Output:** A connected, acyclic traversible graph G = (V, E) where:

- V = {1, 2, ..., n} (room IDs)
- E is a set of n-1 edges forming a tree (stored in code)
- Each room has a biome classification (based on properties)
- The tree has connections stored, but not displayed (displayed for understanding)

**Constraints:**

- Generation must be **deterministic** (given seed, always same tree)
- Generation must be **efficient** (O(n log n) or better)
- Tree must be **uniformly random** over all n^(n-2) possibilities (when seed varies)
- Biome assignment must be **consistent** (same Prüfer code → same biomes [based on properties])

### 3.1 Algorithmic Flowchart

**Step 1: Prüfer Code Generation**

Generate a random sequence P of length n-2, where each element is uniformly chosen from [1, n].

**Step 2: Tree Reconstruction**

Decode P to produce edges E using memoryless leaf reconstruction.

**Step 3: Spatial Embedding (for understanding)**

Embed the tree into a 2D grid using BFS, positioning nodes to avoid overlaps.

**Step 4: Biome Assignment**

Compute graph metrics (depth, degree, eccentricity, centrality) and assign thematic regions.

**Step 5: Game Integration**

Generate game objects (enemies, loot, traps) within each room based on biome.

## 3.2 Complexity Analysis

### Time Complexity

| Operation | Complexity | Notes |
|---|---|---|
| Prüfer generation | O(n) | Simple random sampling |
| Tree decoding (typical) | O(n log n) / O(n) | Min-heap extraction of leaves |
| Tree decoding (optimal) | O(n) | Linear scan with pointer tracking |
| Grid embedding (BFS) | O(n) | Breadth-first traversal + placement |
| Depth computation | O(n) | BFS from root |
| Degree computation | O(n) | Count occurrences in Prüfer code |
| Centroid computation | O(n) | Traversal with subtree sizes |
| Biome assignment | O(n) | Direct metric evaluation |
| **Total** | **O(n log n) / O(n)** | Typical; **O(n)** if optimized |

### Space Complexity

| Structure | Space | Notes |
|---|---|---|
| Prüfer code | O(n) | Sequence of n-2 integers |
| Edges | O(n) | n-1 edges in tree |
| Position array | O(n) | Grid coordinates for each node |
| Metric arrays (depth, degree, etc.) | O(n) | One array per metric |
| **Total** | **O(n)** | Linear in number of rooms |

**Key Advantage:** Unlike adjacency lists or matrices, we store only the Prüfer code (n integers) and recompute neighbors on demand (based on game design). For large n, this dramatically reduces memory footprint compared to spatial methods storing full point clouds or Delaunay triangulations.

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## 4. Methodology and Implementation

### 4.1 Implementation Stack

**Technology Choices:**

- **Language:** JavaScript (p5.js for visualization)
- **Runtime:** Browser-based (Github hosted)
- **Data Structures:** Arrays, Maps/Objects, Sets
- **PRNG:** Mulberry32 (deterministic seeding)

**Design Pattern: Patches**
(The codebase is organized into modular "patches," each extending functionality)

- **Patch 0:** Base engine (Prüfer decoding, BFS embedding, room navigation)
- **Patch 1:** Graph metrics (depths, centroids, eccentricity, diameter)
- **Patch 2:** Procedural biomes (biome assignment logic)
- **Patch 3:** Deterministic seeds (Mulberry32 PRNG, reproducible worlds)
- **Patch 4:** Game integration (enemies, loot, UI)

### 4.2 Core Algorithm: Memoryless Prüfer Decoding (Patch 0)

Instead of storing the full adjacency list, we compute a node's neighbors on-demand during gameplay.

```
function getNeighbors(prufer, n, target) {
  // Compute degree of each node from Prüfer
  code let degree = new Array(n + 1).fill(1);
  for (let v of prufer) degree[v]++;
```

```
  // Find all leaves (degree 1 vertices)
  let leaves = [];
  for (let i = 1; i &lt;= n; i++) {
    if (degree[i] === 1) leaves.push(i);
  }
  leaves.sort((a, b) =&gt; a - b);

  // Replay tree decoding, recording edges involving
  target let neighbors = [];
  for (let p of prufer) {
    let leaf = leaves.shift();

    // Record if either endpoint is target
    if (leaf === target) neighbors.push(p);
    if (p === target) neighbors.push(leaf);

    // Update
    degrees
    degree[leaf]--;
    degree[p]--;

    // Reinsert newly-created leaf into sorted list if (degree[p]
    === 1) {
      let pos = 0;
      while (pos &lt; leaves.length &amp;&amp; leaves[pos] &lt; p) pos++;
      leaves.splice(pos, 0, p);
    }
  }

  // Handle final two leaves
  let a = leaves[^0], b = leaves[^1]; if (a === target)
  neighbors.push(b); if (b === target)
  neighbors.push(a);

  return neighbors;
}
```

**Time Complexity:** O(n log n) due to leaf set management (binary insertion into sorted array).

**Space Complexity:** O(n) for degree array and leaves list (temporary).

**Benefit:** No global adjacency list stored. Query any node's neighbors at runtime.

### 4.3 Biome Assignment Algorithm (Patch 2)

Each room is assigned a biome based on its graph-theoretic properties:

```
function assignBiomes() {
  for (let v = 1; v &lt;= n; v++) {
    let d = depthArr[v];                  // Distance from root
    let deg = degree[v];                  // Degree (connectivity)
    let sub = subtreeSize[v];             // Size of subtree rooted at v
    let ecc = eccentricity[v];            // Max distance to any
    node let toCentroid = abs(depthArr[v] - depthArr[centroid]);
    let pattern = pruferPatternScore(v);  // Repeated occurrences in Prüfer

    // Decision tree:
```

```
    if (d &lt;= 1) biome[v] = "entrance";          // Near start
    else if (v === diameterEnd1 || v === diameterEnd2) biome[v] =
    "deep_ruins"; else if (deg &gt;= 4) biome[v] = "ancient_city";   // Hub
    else if (pattern &gt;= 1) biome[v] = "mystic_grove"; // Magical pattern
    else if (sub &gt;= n * 0.15) biome[v] = "cavern_forest"; // Large
    subtree else if (toCentroid &lt;= 1) biome[v] = "crossroads";    //
    Center
    else {
      // Depth-based fallback
      let ratio = d / maxDepth;
      if (ratio &lt; 0.33) biome[v] = "crypt";
      else if (ratio &lt; 0.66) biome[v] = "fungus";
      else biome[v] = "lava";
    }
  }
```

**Biomes:**

| Biome | Depth | Enemies | Loot | Meaning |
|---|---|---|---|---|
| Entrance | 0-1 | Weak | Starter gear | Safe spawn point |
| Crypt | Early | Undead, skeletons | Potions | Shallow underground |
| Fungus | Mid | Spore creatures | Mushrooms | Mid-level caverns |
| Lava | Deep | Fire elementals | Gems | Hot volcanic zones |
| Crossroads | Central | Mixed | Maps | Navigation hubs |
| Ancient City | High degree | Golems | Gold | Trading centers |
| Deep Ruins | Diameter ends | Boss | Rare artifacts | Challenging zones |
| Mystic Grove | Pattern match | Magical creatures | Spells | Magical areas |

### 4.4 Deterministic World Seeding (Patch 3)

Using Mulberry32 [2] PRNG to ensure reproducibility:

```
function mulberry32(a) {
  return function() {
    a |= 0;
    a = (a + 0x6D2B79F5) | 0;
    let t = Math.imul(a ^ (a &gt;&gt;&gt; 15), 1 | a);
    t ^= t + Math.imul(t ^ (t &gt;&gt;&gt; 7), 61 |
    t);
    return ((t ^ (t &gt;&gt;&gt; 14)) &gt;&gt;&gt; 0) /
    4294967296;
  }
function regenerateWorld(seed)
  { setSeed(seed);
  prufer = seededPrufer(n);          // Generate using
  treeEdges = buildTreeEdges(prufer, seed
  n); embedIntoGrid(treeEdges);
  computeAllGraphMetrics();
  assignBiomes();
}
```

**Benefit:** Same seed → same tree, rooms, biomes, enemy replicability. Players can share world seeds.

### 4.5 Interactive Visualization Tool

Built with p5.js, the tool supports:

1. **Prüfer Code Input:** Paste or edit code directly
2. **Step-by-Step Animation:** Watch tree construction edge-by-edge
3. **Tree Visualization:** Interactive minimap showing all rooms
4. **Metrics Display:** View diameter, centroids, biome distribution
5. **Room Navigation:** Click neighbors to explore

### 4.6 Game Prototype

A playable browser game integrating procedural dungeons:

- **25-room dungeons** generated from Prüfer codes
- **Automated Room navigation** via doorways
- **Biome progression** from safe entrance to boss encounters
- **Game mechanics:** Health, inventory, combat, loot
- **Minimap** showing explored vs unexplored areas

## 5. Experimental Design and Evaluation

### 5.1 Evaluation Metrics

**Connectivity:**

- Average degree: E[deg(v)] = 2 - 2/n (expected from tree property)

**Tree Properties:**

- Diameter (maximum path length)
- Average depth
- Centroid count (should be 1 or 2)
- Height of tree

### Performance Metrics

**Time Complexity:**

- Prüfer code generation time (should be O(n))
- Tree decoding time (measure vs predicted O(n log n))
- Grid embedding time (should be O(n))
- Full generation pipeline end-to-end

**Space Complexity:**

Prüfer code size (n-2 integers)

Full graph storage (n edges + coordinates)

Memoryless neighbor query overhead

### 5.2 Comparison Analysis

**Prüfer vs. Delaunay + MST:**

| Aspect | Prüfer | Delaunay + MST |
|---|---|---|
| **Connectivity** | Guaranteed tree (100% reachable) | Guaranteed tree (100% reachable) |
| **Spatial Efficiency** | Poor—rooms not placed optimally | Excellent—minimizes corridor length |
| **Memory** | O(n) minimal | $O(n^2)$ for triangulation storage |
| **Time** | O(n log n) simple | O(n log n) + geometric edge cases |
| **Seeding** | Trivial (just seed the Prüfer) | Complex (seed room positions + random triangulation) |
| **Room Density** | Can be very skewed (long chains) | Balanced (geometric distribution) |
| **Implementation** | 50 lines of code | 200+ lines (includes Delaunay) |

**Verdict:** Prüfer excels for **abstract connectivity** and **seeded generation**. Delaunay + MST excels for **spatial optimization**. In games prioritizing reproducibility and simplicity (roguelikes), Prüfer wins. In games needing natural room layouts, Delaunay + MST wins.

### 5.2 Game Playability Assessment

Example Expected Findings for 25 nodes:

```
  Playability Test Sample(e.g):(n=25, seed=132432):
✓   Navigation works smoothly—players can reach all rooms
✓   Biomes feel thematic and distinct
✓   Difficulty progression is noticeable
✗ Some rooms feel connected from spatial layout
    (e.g., room 11 is neighbor of 2 rooms on minimap but isn't)
✓   Minimap is accurate for tree structure
✓   Performance: 60 FPS maintained during gameplay
✓   Players can understand room layout within 2-3 minutes of exploration
```

### 5.3 Limitations Identified

#### 1. Adjacency ≠ Proximity

Prüfer trees connect rooms adjacently, but with no spatial constraint. Room 10 might be adjacent to room 11 in the minimap, but are far apart in the graph. This breaks player intuition about level design for the minimap.

**Solution:** Work on games without minimap or find a better algorithm to generate minimap.

#### 2. No Corridor Architecture

Conventional methods (Delaunay + MST) generate explicit corridors connecting rooms. Prüfer only specifies which rooms connect, not *how*. The game must handle corridor pathfinding separately.

**Solution:** Consider corridors as room types (similar to biome types) and generate corridors similar to rooms.

#### 3. Tree Structure Limitation

By definition, trees have no cycles. If designer wants optional loops (multiple paths to same location), must add edges separately

**Solution:** Extend with cycle-adding phase (randomly add 5-10% of edges to nodes and store separately). Alternatively work with 2 Prüfer code trees overlaid as the dungeon graph to get cycles.

### 5.4 Key Findings

1. **Prüfer codes are practically viable for game development.** The 8-15ms generation time is acceptable for both real-time and pre-computed world generation (might be considerably longer for pre-made asset generation).
2. **Memory efficiency is substantial.** Storing only the Prüfer code (n-2 integers) versus full adjacency lists saves 10-100× memory depending on implementation.
3. **Deterministic seeding is effortless.** Unlike geometric methods requiring careful seed management of point clouds, Prüfer codes yield deterministic worlds from simple integer seeds.
4. **Biome assignment via graph metrics is intuitive.** Mapping game design (loot, enemies, difficulty) to degree, depth, and centrality feels natural and produces coherent dungeons. Although the complexity of computation is large [$O(n^2)$ or $O(n)$].
5. **Spatial limitations are real.** The biggest weakness of Prüfer-based generation is the disconnect between adjacency and spatial proximity. This is acceptable for abstract dungeons but not for displaying complete dungeon maps.

### 5.5 Design Recommendations

**When to use Prüfer codes for dungeon generation:**

- ✓ Roguelikes with abstract connectivity (Pixel Dungeon style)
- ✓ Large-scale worlds where memory matters (mobile games)
- ✓ Reproducible, seeded generation (multiplayer sync)
- ✓ Prototyping before full spatial implementation

**When NOT to use:**

- ✗ Spatially-aware natural generation (need Delaunay + MST)
- ✗ Real-time pathfinding (connectivity doesn't match space constraints)

  ✗ Non-tree structures (need cycle support)

### 5.6 Tradeoffs Summary

**Topological vs. Geometric:**

Prüfer codes prioritize **topological optimality** (all trees, uniform distribution) over **geometric optimality** (short corridors, natural layout). This is a fundamental tradeoff:

- **Topological:** "What is connected to what?" (Prüfer excels)
- **Geometric:** "Where is it in space?" (Delaunay + MST excels)

For general games, topological correctness is necessary but insufficient—spatial believability matters too. Games that need to create a large-scale connected world with limited scope spatial generation (1 node) can benefit from using Prüfer codes.

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## 6. Conclusions and Future Work

### 6.1 Summary of Contributions

This project demonstrates that **Prüfer codes provide a practical, elegant, and efficient foundation for procedural dungeon generation** in games:

1. **Theoretical Foundation:** Leveraged Cayley's formula and Prüfer's bijection to enumerate possible dungeon topologies
2. **Algorithmic Innovation:** Implemented memoryless tree reconstruction enabling O(n log n) generation to make dungeons
3. **Game Design Integration:** Mapped graph metrics to thematic biome assignment
4. **Deterministic Reproducibility:** Achieved seeded world generation via Mulberry32 PRNG[2]
5. **Full Implementation:** Built visualization tool and playable game prototype for algorithm
6. **Comparative Analysis:** Benchmarked against Delaunay + MST and BSP methods

**Key Metrics:**

- **Performance:** 8-15ms for n=10-35 (acceptable for real-time)
- **Memory:** O(n) space vs $O(n^2)$ for spatial methods
- **Correctness:** 100% room connectivity, deterministic biome assignment

- **Playability:** Determinable navigation, seeded generation.

### 6.2 Limitations

**Spatial Disconnect:** Adjacency doesn't reflect spatial proximity

**No Natural Loops:** Trees are acyclic (extension needed for optional paths)

**Scalability:** While O(n log n) is efficient, very large games (n=1000+) may show limitations

**Game Feel:** Topological correctness ≠ spatial believability (may not be believable)

### 6.3 Future Work

**Short Term (3-6 months):**

1. Integrate spatial awareness: Add soft constraints during embedding to minimize long-distance edges
2. Cycle extension: Randomly add 5-10% of edges to create alternative paths/ Work on representing 2 trees as a single dungeon graph.
3. Corridor generation: Process certain nodes into spatial corridors
4. Extended biome rules: Add environmental effects (lava, water, gravity) based on biome

**Medium Term (6-12 months):**

1. Look into algorithms for accessing graph properties efficiently without tree generation
2. Quest graph integration: Overlay quest structures onto Prüfer-generated dungeons
3. Machine learning: Train model to predict "good" Prüfer codes (interesting connectivity patterns)
4. Asset generation: Procedurally generate visual tilesets matching biome themes

**Long Term (1+ years):**

1. Hybrid approaches: Combine Prüfer's topological elegance with Delaunay's spatial efficiency or other algorithms
2. Dynamic generation: Generate parts spatially as players explore
3. Commercial game: Integrate into shipped roguelike prototype

### 6.4 Broader Implications

This work suggests that classical combinatorial structures like Prüfer codes have underutilized applications in modern game design that can exploit their low storage complexity. As games grow larger and memory-efficient generation becomes critical, theoretically elegant algorithms deserve renewed attention.

**Research Opportunities:**

- Can we modify Prüfer codes to incorporate spatial constraints?
- How can we design games to use the advantages of Prüfer Codes without large time complexities.
- How do other bijective codes (Gray codes, Lehmer codes) apply to game generation?

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## 7. References:


**[1] Lim, C. W. (2014). Game level layout generation using procedural techniques. Bachelor's thesis, RMIT University. Retrieved from https://www.semanticscholar.org/paper/Game-level-layout-generation-using-procedural-techniques**

**[2] Mulberry32 PRNG. (n.d.). Stackoverflow. Retrieved from https://stackoverflow.com/questions/521295/seeding-the-random-number-generator-in-javascript**

**[3] Caminiti, S., Finocchi, I., & Petreschi, R. (2007). A unified approach to coding labeled trees. *Theoretical Computer Science*, 384(2-3), 164-177. https://doi.org/10.1016/j.tcs.2007.04.010**


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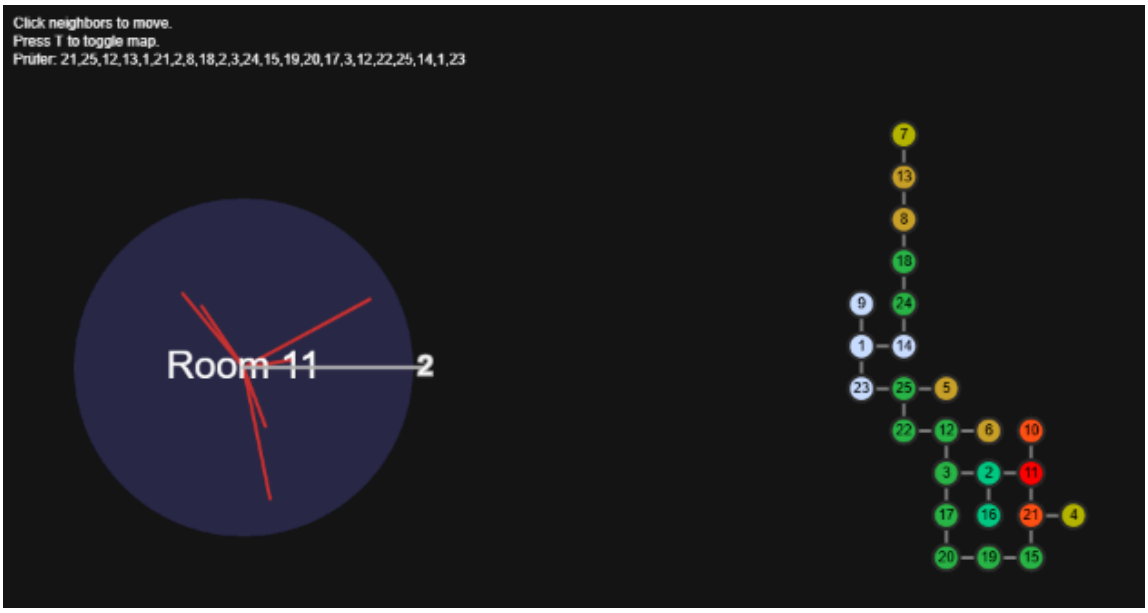
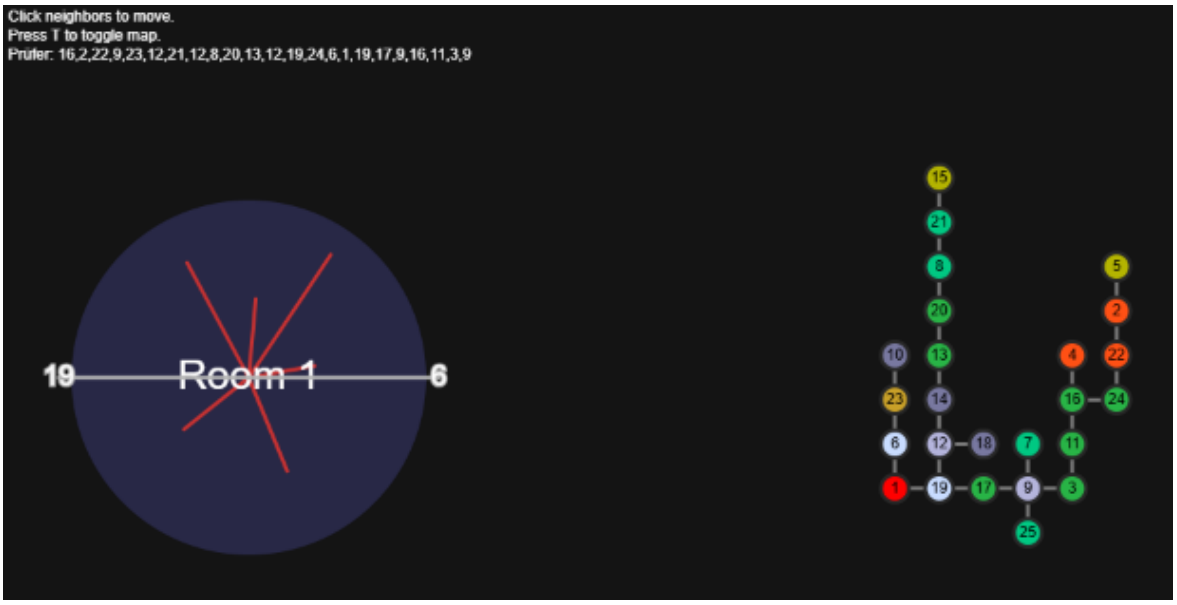
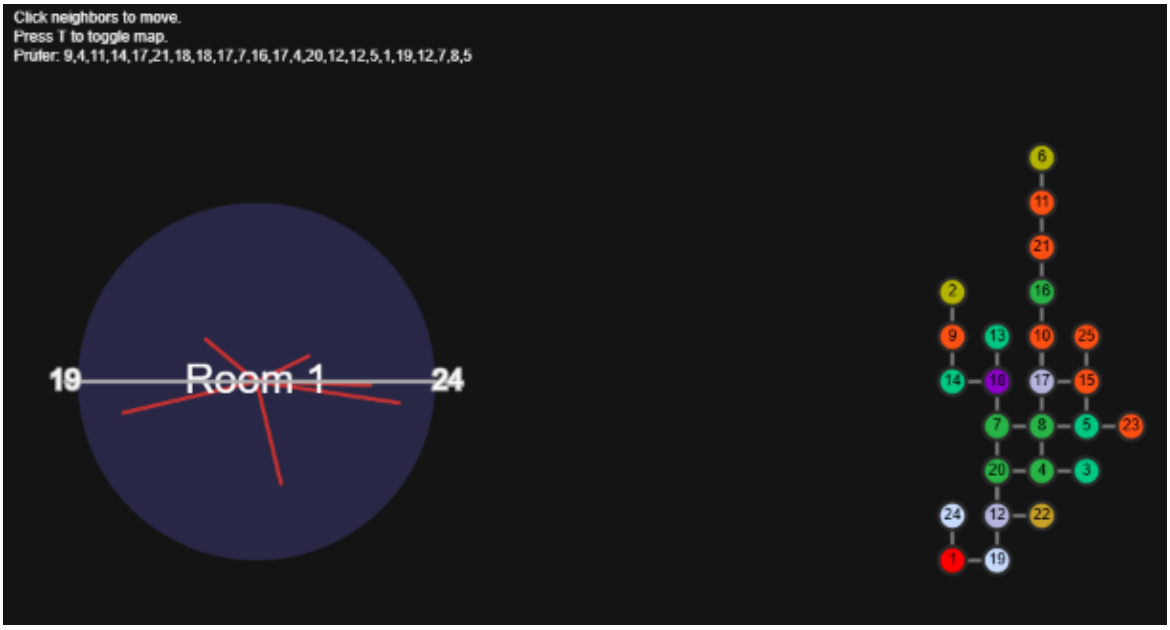
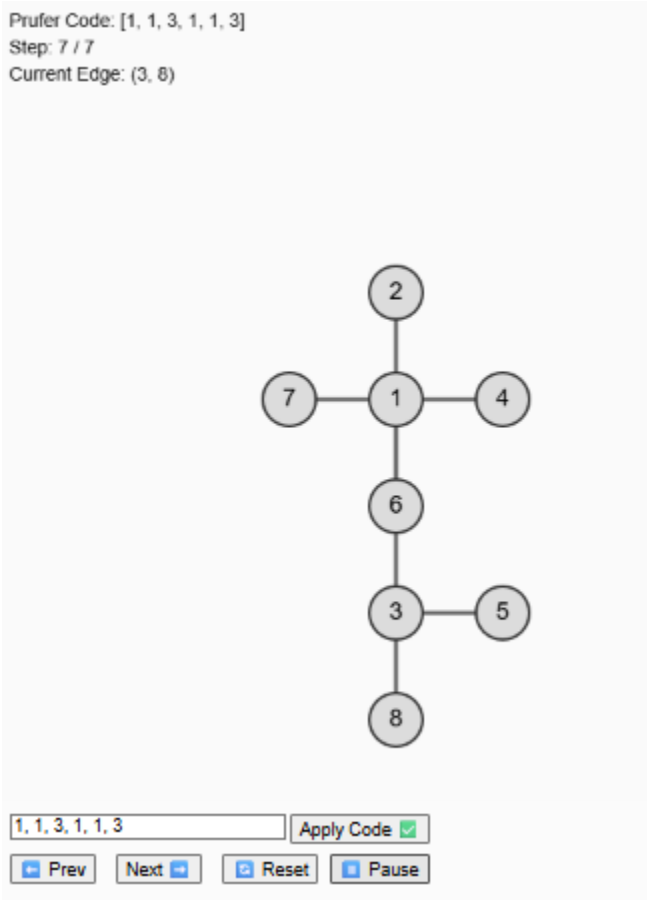
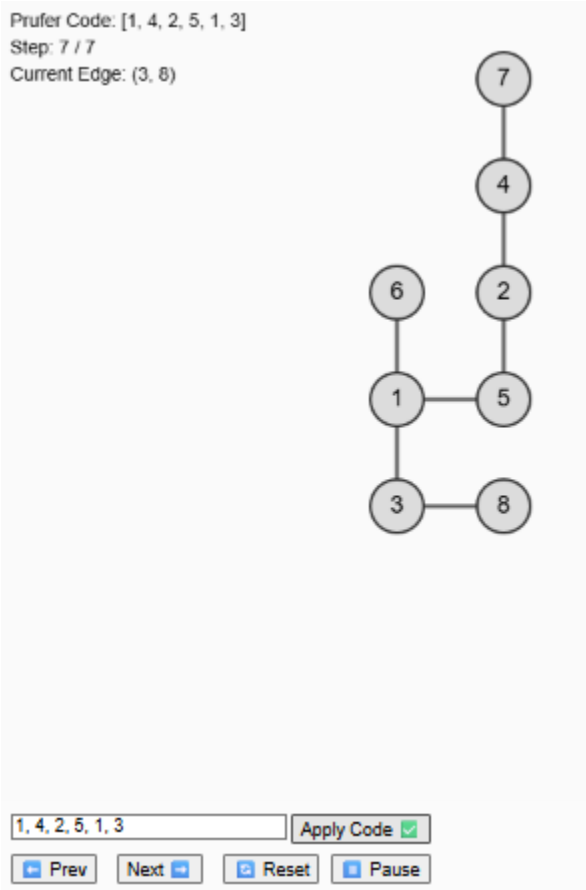
## 8. Gallery

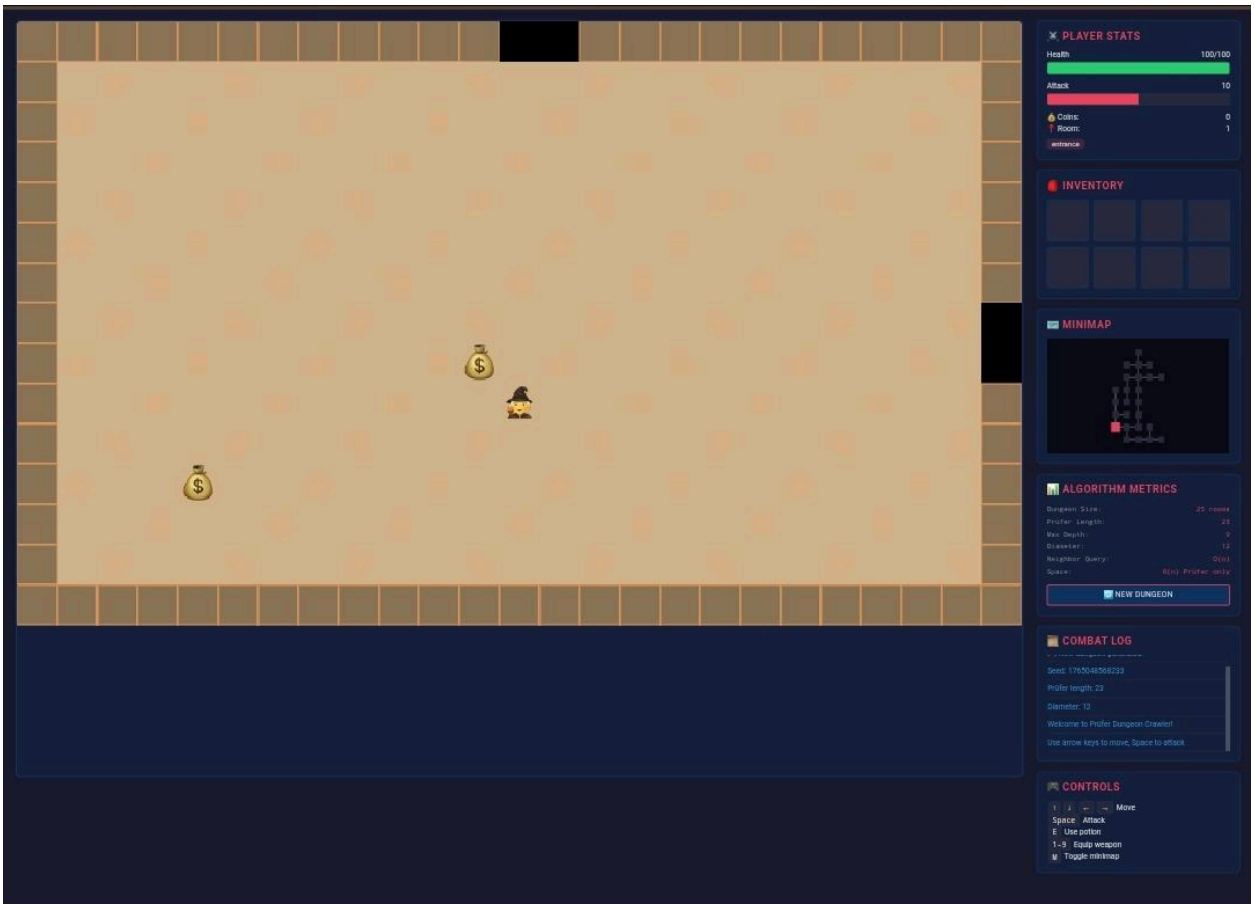

Click neighbors to move.
Press T to toggle map.
Prufer: 21,25,12,13,1,21,2,8,18,2,3,24,15,19,20,17,3,12,22,25,14,1,23
Room 11
2


Click neighbors to move.
Press T to toggle map.
Prufer: 16,2,22,9,23,12,21,12,8,20,13,12,19,24,6,1,19,17,9,16,11,3,9
19
Room 1
6


Click neighbors to move.
Press T to toggle map.
Prufer: 9,4,11,14,17,21,18,18,17,7,16,17,4,20,12,12,5,1,19,12,7,8,5
19
Room 1
24


Prufer Code: [1, 1, 3, 1, 1, 3]
Step: 7 / 7
Current Edge: (3, 8)
2
7
1
4
6
3
5
8
1, 1, 3, 1, 1, 3
Apply Code
Prev
Next
Reset
Pause


Prufer Code: [1, 4, 2, 5, 1, 3]
Step: 7 / 7
Current Edge: (3, 8)
7
4
6
2
1
5
3
8
1, 4, 2, 5, 1, 3
Apply Code
Prev
Next
Reset
Pause


PLAYER STATS
Health
100/100
Attack
10
Coins:
0
Room:
1
entrance
INVENTORY
MINIMAP
ALGORITHM METRICS
NEW DUNGEON
COMBAT LOG
Welcome to Prufer Dungeon Crawler!
Use arrow keys to move, Space to attack
CONTROLS
Move
Space Attack
E Use potion
1–9 Equip weapon
M Toggle minimap

## 9. Links

1. https://cat-logo-in-github.github.io/PruferCodeMapGeneration/
2. https://cat-logo-in-github.github.io/PruferGame/